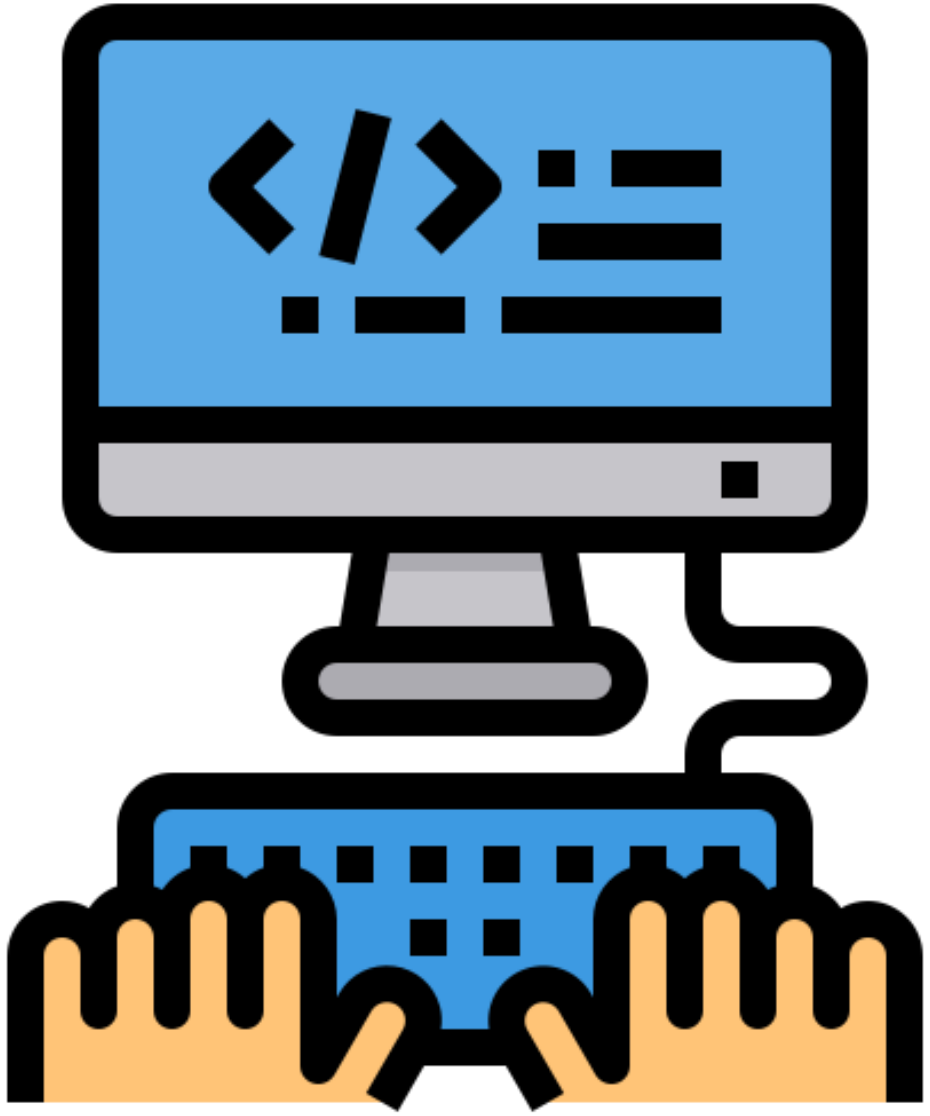
CS 24 AP Computer Science A Review

Week 6: Array

DR. ERIC CHOU
IEEE SENIOR MEMBER

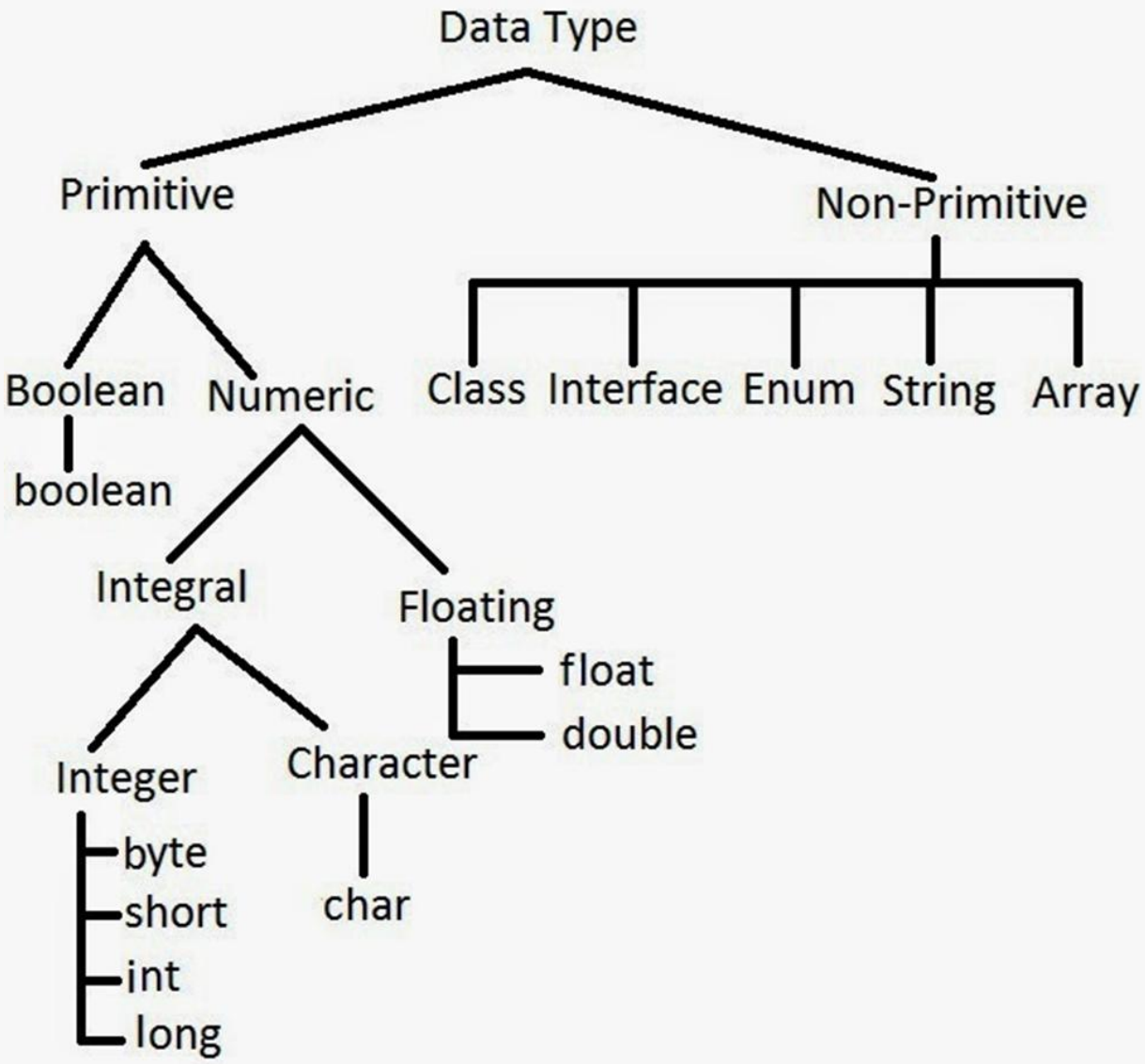
Topics

- Reference Data Type
- Array Definition
- Basic Array Class
- For Each Loop
- Array Processing
- Basic Array Algorithms

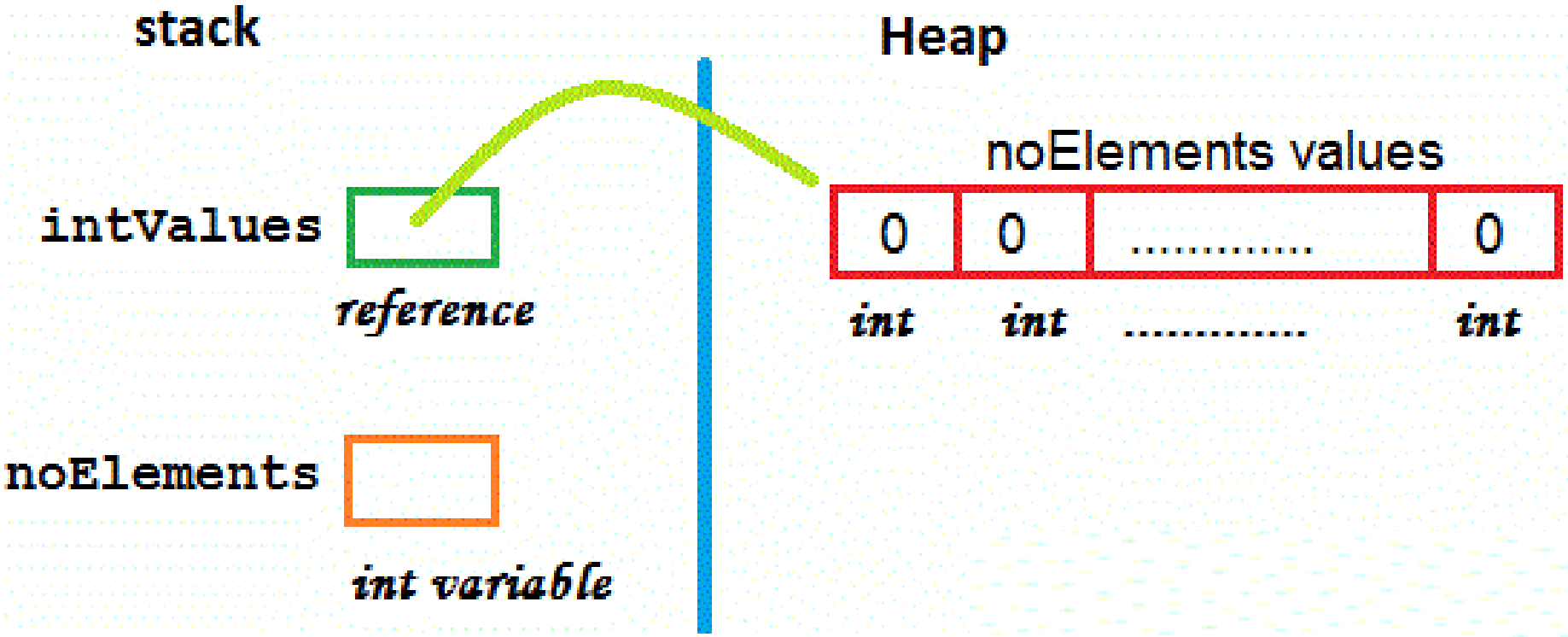


Reference Data Type

SECTION 1


Data Type
Primitive
Non-Primitive
Boolean
Numeric
Class
Interface
Enum
String
Array
boolean
Integral
Floating
float
double
Integer
Character
byte
short
int
long
char

Reference Date Type

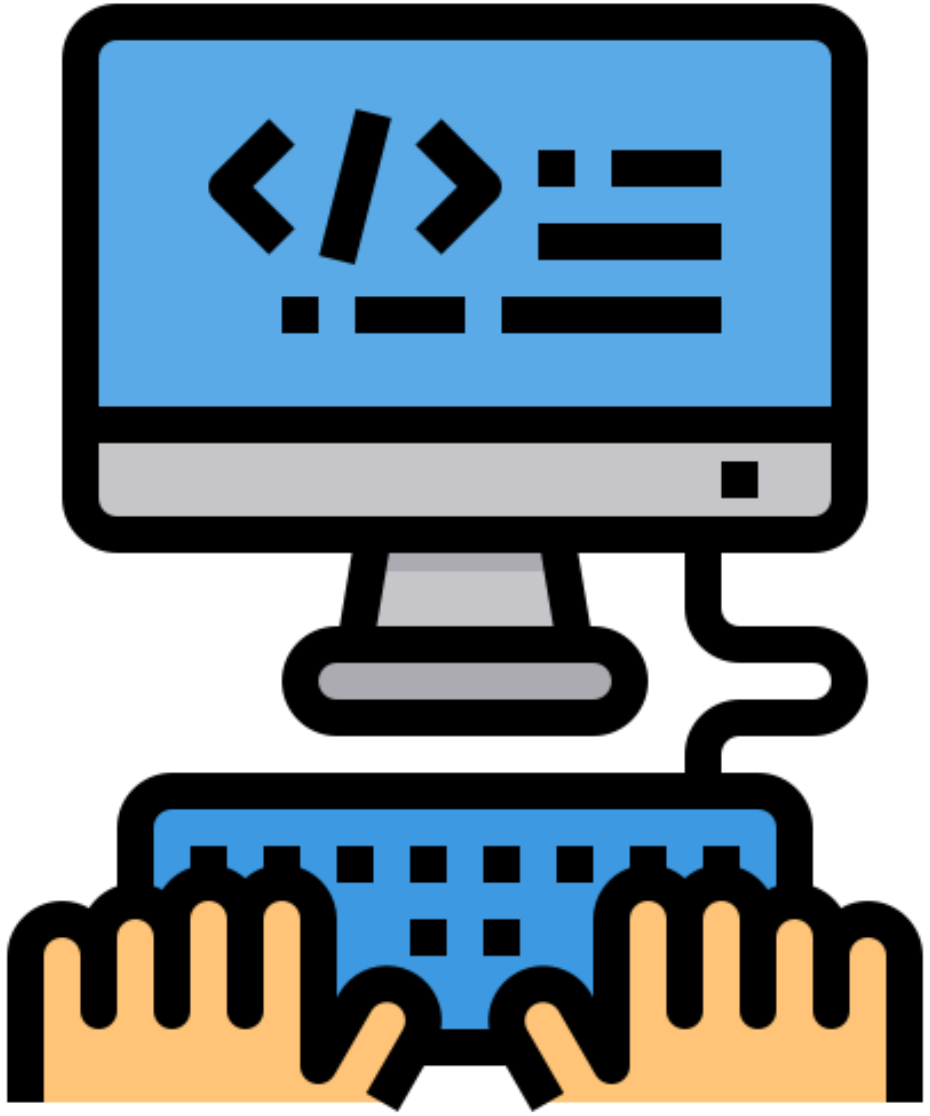


Three Important Complex Data Structures

There are five basic data structures (ADT) that are tested on the AP Computer Science A Exam:

- String (Covered in Week 2)
- The array (of one-dimensional array)
- The 2-D array
- The ArrayList
- Class (Structure, Data Record)

Situation	Recommendation	Justification
A program helps an elevator know what floor it is on.	Array	The number of floors is fixed. The elevator can go to any floor by knowing what number it is.
Facebook keeps track of how many friends you have.	ArrayList	The number of friends you have on Facebook may increase or decrease. You are allowed to add or remove anyone at any time regardless of where they are in the list.
Your program is going to simulate chess, Candy Crush, or 2048	2-D Array	Each of these games can be simulated on either a square or rectangular grid in which the row and column are used to find out what is in each cell.
A cell phone keeps track of text messages	ArrayList	The number of text message on a cell phone can increase or decrease. You can even delete all of the messages.
A program keeps track of classes you have each period of the school day.	Array	The number of class periods in the school day is fixed. Each period is assigned a value

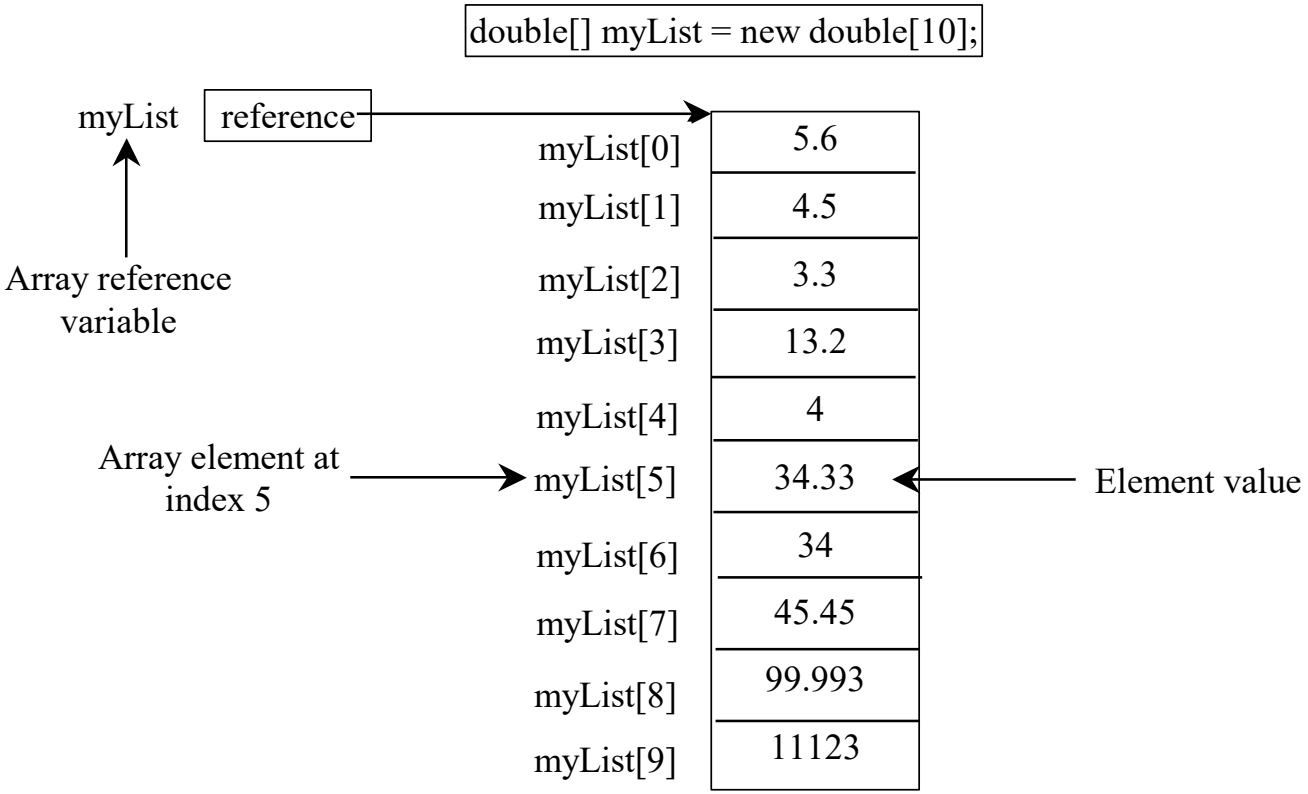


Array Definition

SECTION 2

Introducing Arrays

- Array is a data structure that represents a collection of the same types of data.



Pre-defined Array

- **Pre-defined Array:**

```
int[] intArray = {1, 2, 3, 4, 5, 6, 7, 8, 9, 0};
```

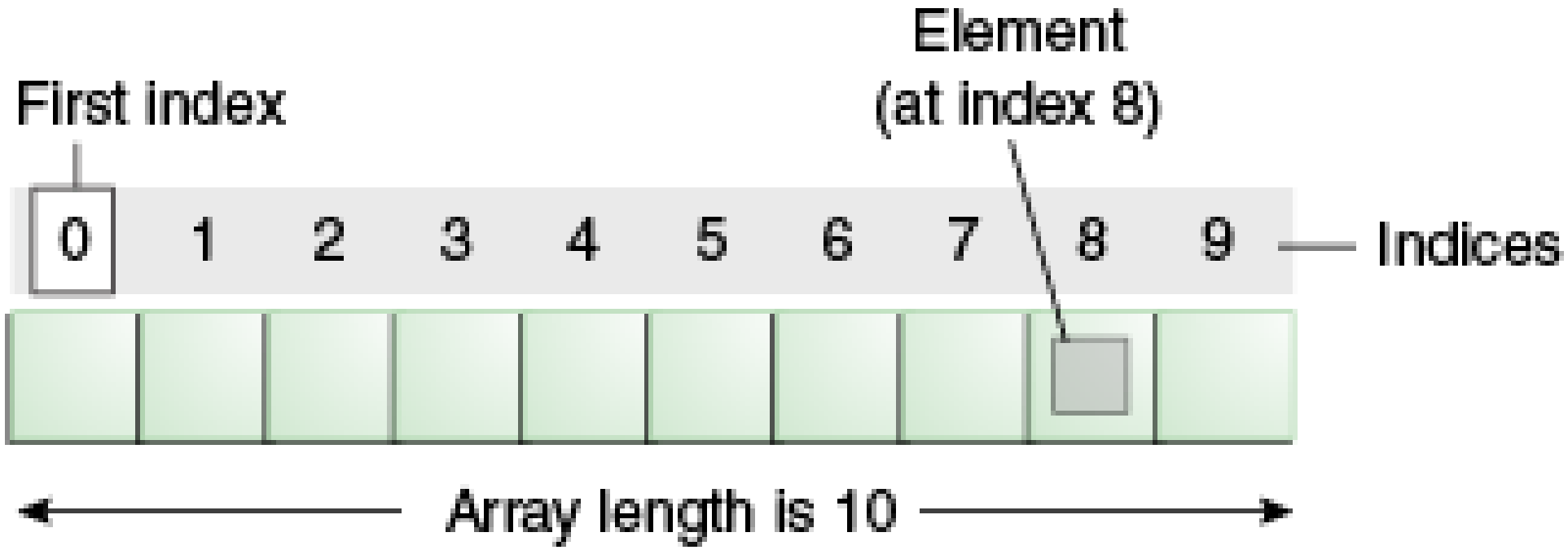
- **Array Declaration with array memory allocation:**

```
int[] intArray2 = new int[10];
```

- **Array declaration and data assignment with anonymous array:**

```
int[] intArray3;
        intArray3 = new int[]{1, 2, 3, 4, 5, 6, 7, 8, 9, 0};
```

array.length



The **new** Operator

- An object of a class is named or declared by a variable of the class type:

  ```
  ClassName classVar;
  ```

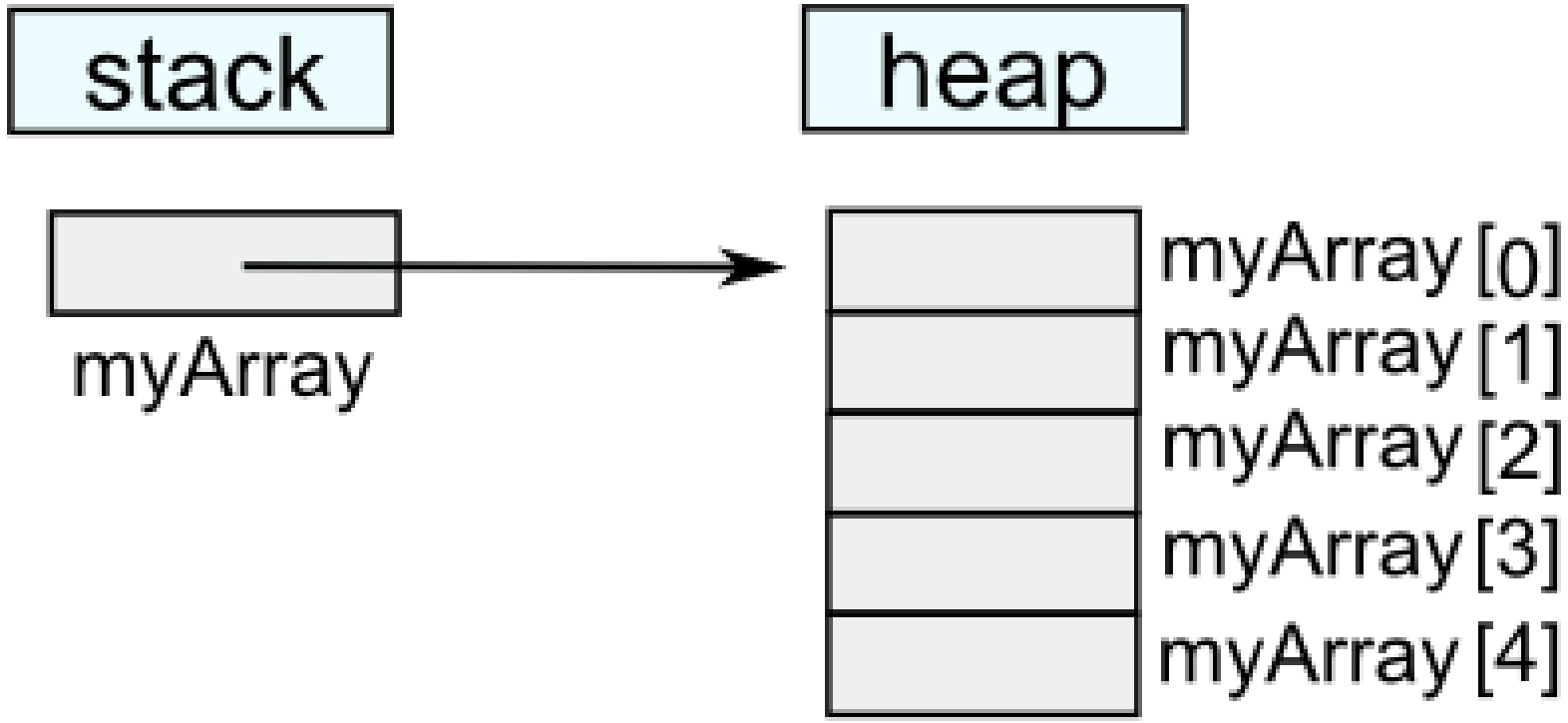
- The new operator must when be used to create the object and associate it with its variable:

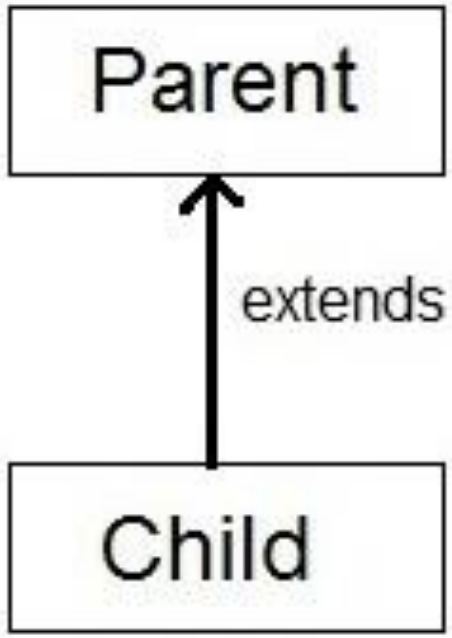
  ```
  classVar = new ClassName();
  ```

- These can be combined as follows:

  ```
  className classVar = new  ClassName();
  ```

new Operator






```
Parent p = new Child( );
```

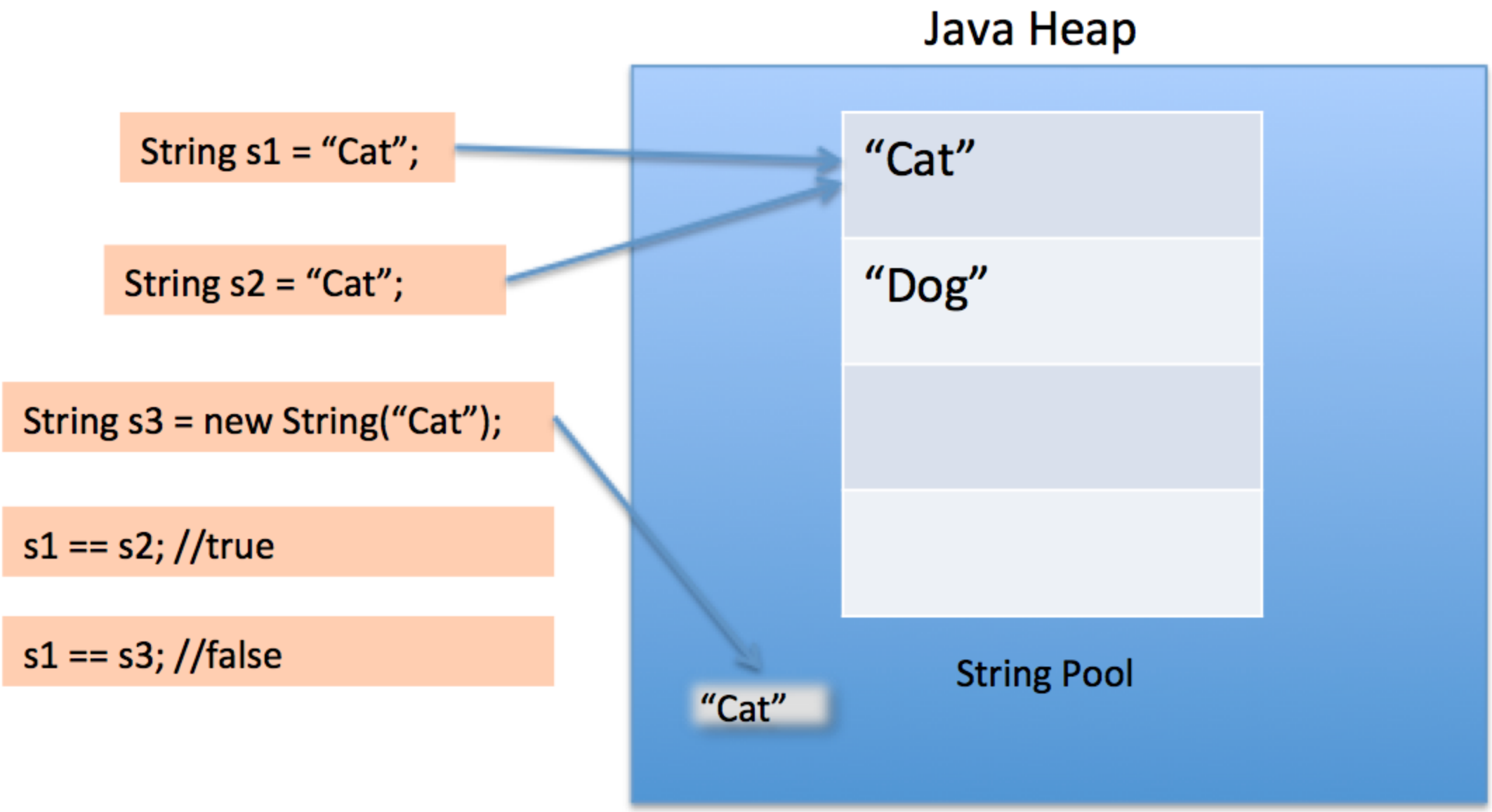
Upcasting

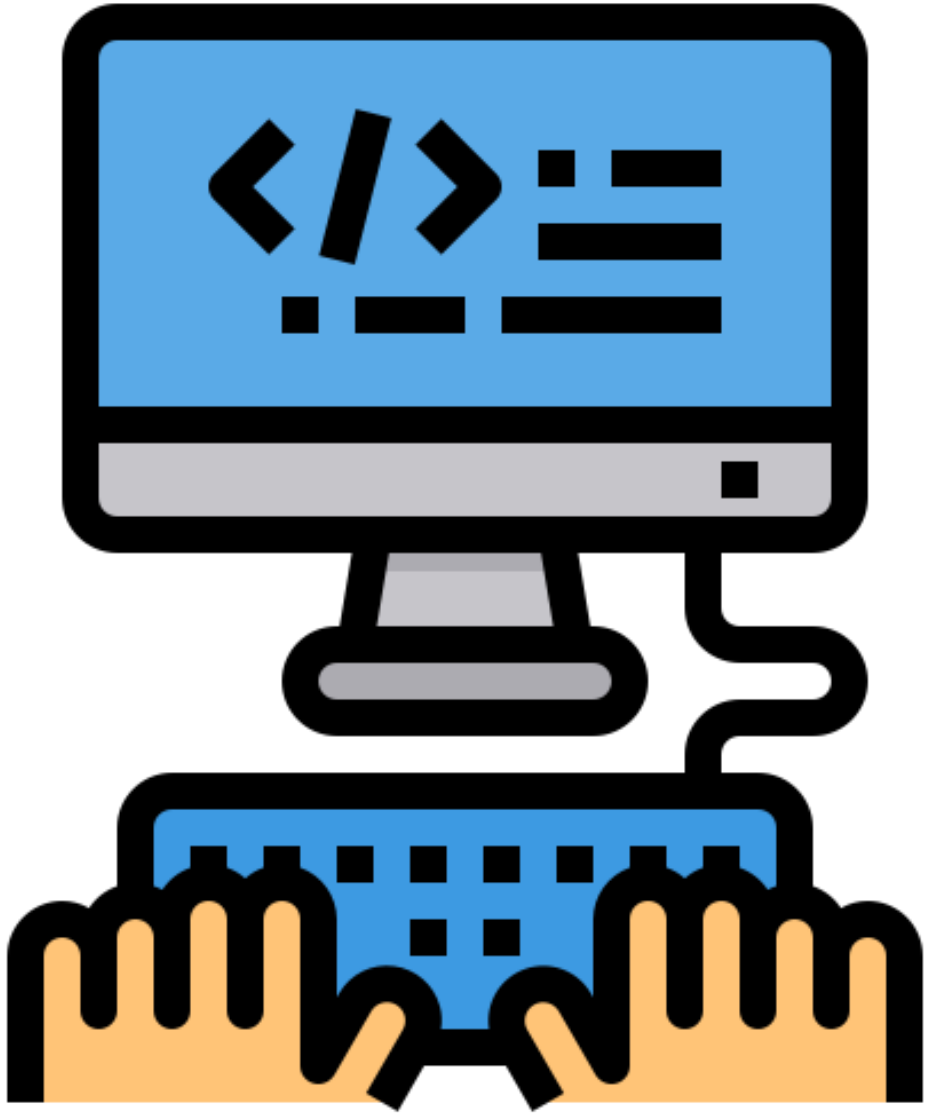
~~Child c = new Parent( );~~

Compile time error

```
Child c = ( Child ) new Parent( );
```

Downcasting but throws ClassCastException at runtime.


Java Heap
String s1 = “Cat”;
String s2 = “Cat”;
String s3 = new String(“Cat”);
s1 == s2; //true
s1 == s3; //false
“Cat”
“Dog”
“Cat”
String Pool



Basic Array Class

SECTION 3

create an array with random values	`double[] a = new double[n];`
	`for (int i = 0; i < n; i++)`
	`  a[i] = Math.random();`

print the array values, one per line	`for (int i = 0; i < n; i++)` `  System.out.println(a[i]);`

find the maximum of the array values	`double max = Double.NEGATIVE_INFINITY;`
	`for (int i = 0; i < n; i++)`
	`  if (a[i] > max) max = a[i];`

compute the average of the array values	`double sum = 0.0;`
	`for (int i = 0; i < n; i++)`
	`  sum += a[i];`
	`double average = sum / n;`

reverse the values within an array	`for (int i = 0; i < n/2; i++)`
	`{`
	`  double temp = a[i];`
	`  a[i] = a[n-1-i];`
	`  a[n-i-1] = temp;`
	`}`

copy sequence of values to another array	`double[] b = new double[n];`
	`for (int i = 0; i < n; i++)`
	`  b[i] = a[i];`

eC Learning Channel

Array as Stack
Demo Program: StackOfIntegers.java

Go BlueJ!!!

- **`size`** is also used as the top pointer for the stack.

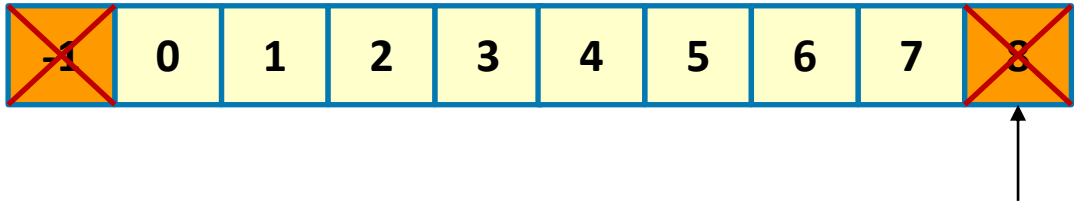
Discrete Space

(last included, end not included)

Count = lastIndex-firstIndex+1 ;

Count = endIndex-startIndex;

Array with 8 elements (indexed from 0 to 7)



8-0 (=7-0+1) = 7- (-1)

Number of elements is 8-0 or 7 – (-1).

Use 7 – (-1) (Last element – the element before first element)

Or, 8-0 (The empty element – the first element)

Base and local index

n-i	-8	-7	-6	-5	-4	-3	-2	-1	0
b+i	4	5	6	7	8	9	10	11	~~12~~
i	0	1	2	3	4	5	6	7	8

Index = Base + local_Index

= b+i

Index = Length - local_neg_Index

= n-i

Base = b = 4, Length = n = $a.length$

Indexing for ASCII code

- Total number of ASCII letters: ‘Z’-’A’+1 (alphabet size)
- Indexing for a letter (‘B’): ‘A’ + (‘B’-’A’)
- Traversing through the whole alphabet: ‘A’ + i; i is from 0 to 25
- Letter indexing is also used for histogram: a[‘X’-’A’] to store the number of occurrence for the letter ‘X’

Sum up an array of 8 elements with index from the last element to the first.

- Write a program to sum up

int a = {1, 2, 3, 4, 5, 6, 7, 8};

- starting from 8 down to 1 with proper indexing.

```
public static void main(String[] args){
    int[] a = {1, 2, 3, 4, 5, 6, 7, 8};

    int sum=0;
    for (int i=a.length-1; i>=0; i--){
        sum+= a[i];
     }
    System.out.println(Arrays.toString(a)+"'s sum="+sum);
 }
```

CopyTo(src, dest, fromIndex) Method

Demo Program: CopyTo.java

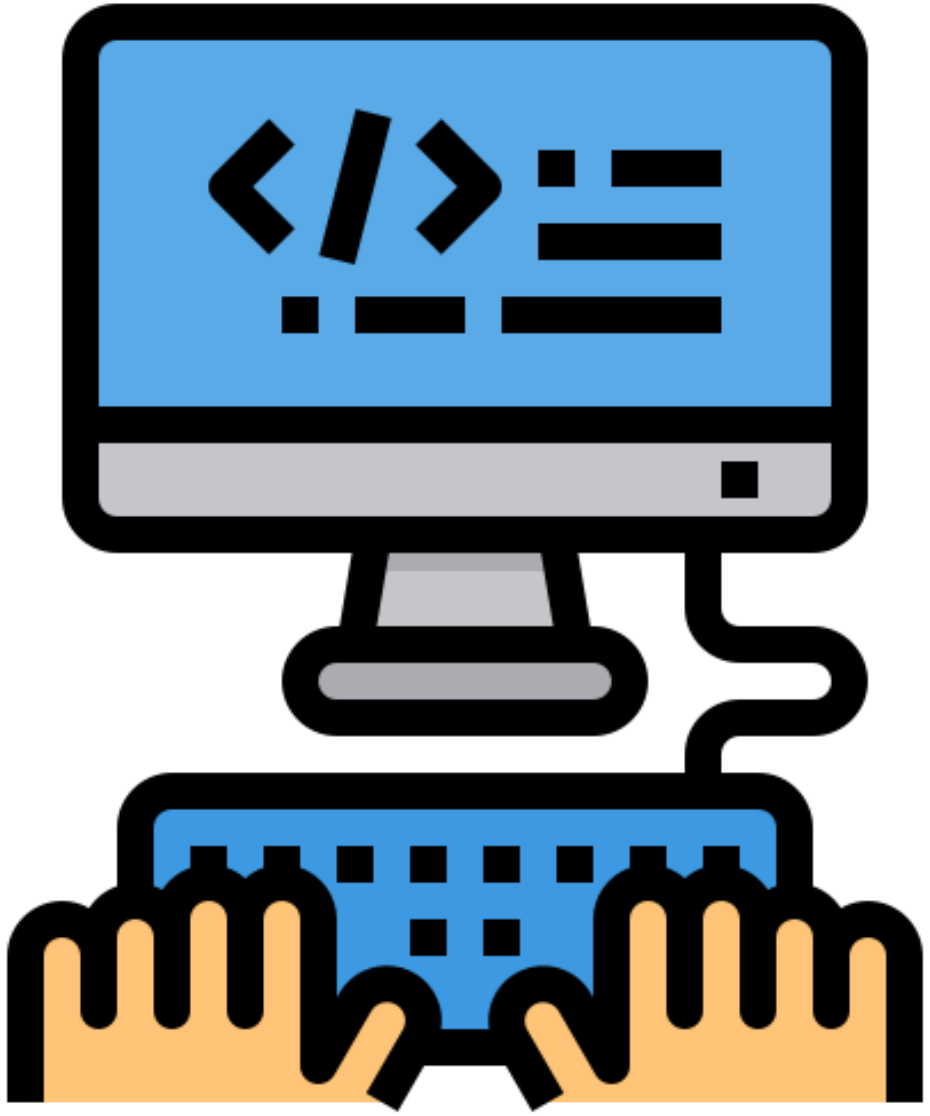
```
static int[] a = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};
static int[] b = {10, 11, 12};
```

```
public static void copyTo(int[] source, int[] destination, int fromIndex){
    if (fromIndex+source.length > destination.length) return;
    for (int i=0; i<source.length; i++){
        destination[fromIndex+i] = source[i];
    }
}
```

Key Methods in Arrays

Key Methods in java.util.Arrays	Descriptions
static List **asList**(T[])	Convert an array to a List (and bind them)
static int **binarySearch**(Object[], key) static int **binarySearch**(primitive[], key)	Search a sorted array for a given value, return an index or insertion point
static int **binarySearch**(T[], key, Comparator)	Search a Comparator-sorted array for a value
static boolean **equals**(Object[], Object[]) static boolean **equals**(primitive[], primitive[])	Compare two arrays to determine if their contents are equal
public static void **sort**(Object[ ]) public static void **sort**(primitive[ ])	Sort the elements of an array by natural order
public static void **sort**(T[], Comparator)	Sort the elements of an array using a Comparator
public static String **toString**(Object[]) public static String **toString**(primitive[])	Create a String containing the contents of an array

(Non-AP: Use with cautions)



For Each Loop

SECTION 4

Enhanced for-loop

Demo Program: ForEach1.java

General Form for the Enhanced for loop (the for-each loop):

```
datatype[] arrayName = /* array filled in some way */
for (datatype temporaryVarible: arrayName){
  // instructions that sue temporaryVariable
}
```

What data structure can be accessed by for-each loop?

Any container class that implements both Iterable Interface and Iterator Interface can be accessed by for-each loop.

Iterable Interface:

iterator() method

Iterator Interface:

hasNext() method

next() method

remove() method

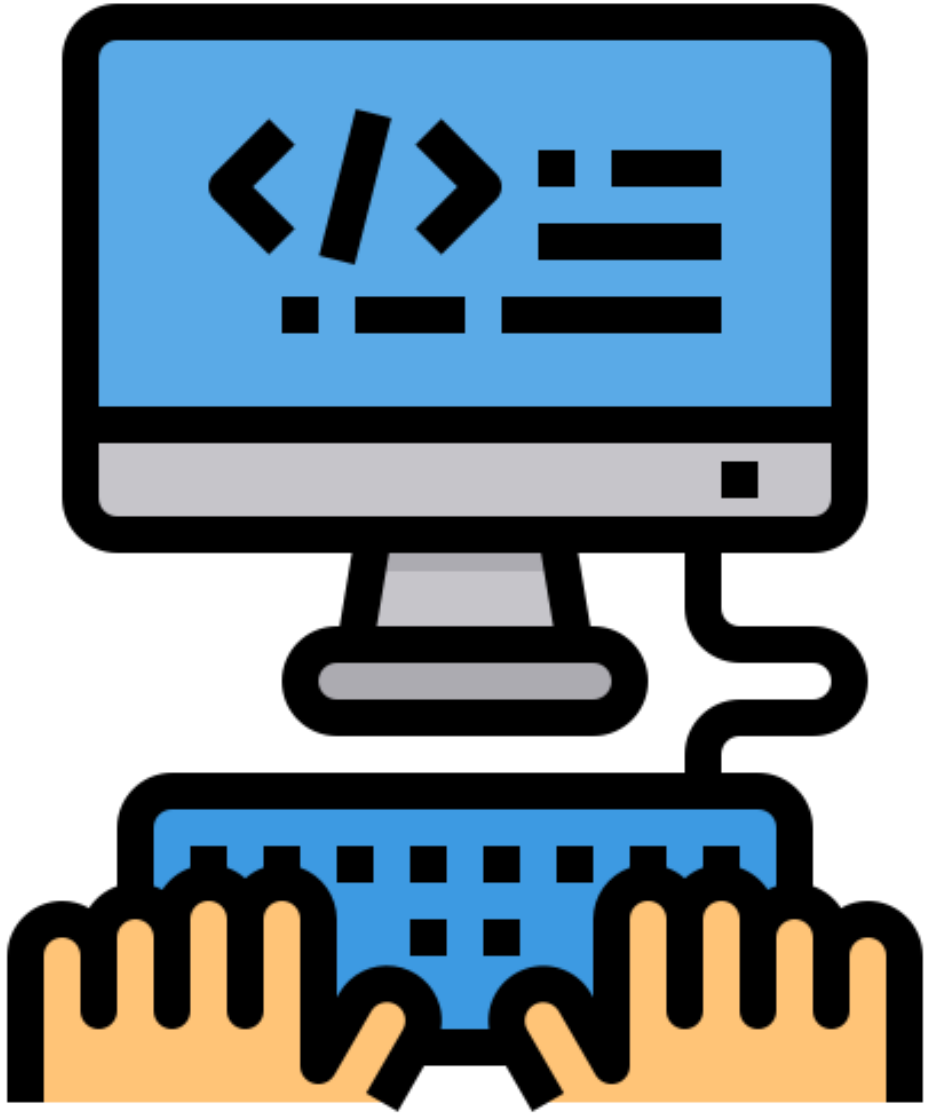
```java
import java.lang.Iterable;
import java.util.Iterator;
import java.util.NoSuchElementException;
import java.util.Arrays;
public class StringList implements Iterable<String>, Iterator<String>{
    String[] list = new String[10];
    int length=0;
    int i = 0;
    StringList(String[] source){
        list = source;
        length = list.length;
     }
    public int size(){ return length; }
    public Iterator<String> iterator(){
        return this;
     }
    public String toString(){
       return Arrays.toString(list);
    }
    public boolean hasNext(){
        return i<length;
    }
    public String next(){
         if (!hasNext()) throw new NoSuchElementException();
         return list[i++];
    }
    public void remove(){
      throw new UnsupportedOperationException();
    }
    public void reset(){
        i=0;
    }
}
```

```java
public static void main(String[] args){
    String[] s = {"alpha", "beta", "gamma", "delta", "epsilon"};
    StringList alist = new StringList(s);
    //System.out.println(alist);
    //System.out.println(alist.size());
    for (String str: alist){
        System.out.println(str);
     }
    for (String str: alist){
        System.out.println(str);
     }
 }
```

BlueJ: Terminal Window - Week3

Options

```
alpha
beta
gamma
delta
epsilon
alpha
beta
gamma
delta
epsilon
```

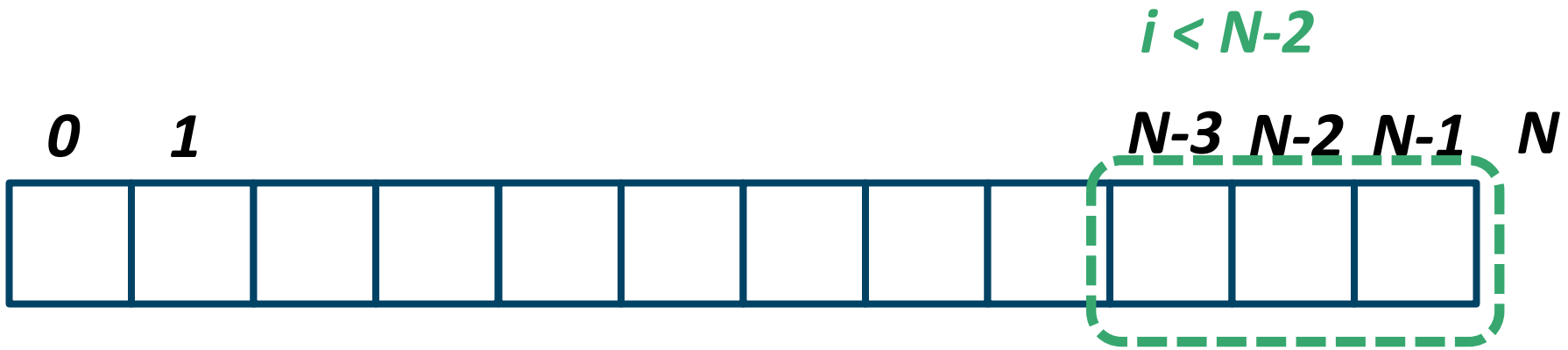


Array Processing

SECTION 5

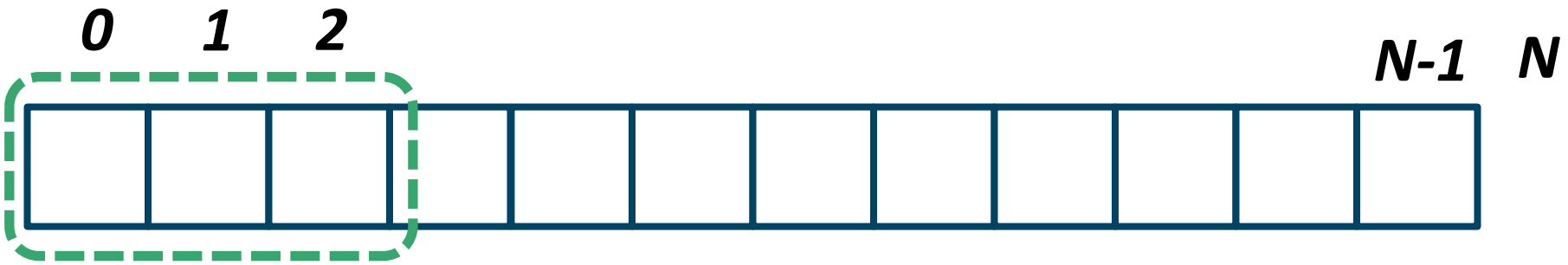
Window of 3

```
for (int i=0; i<arr.length-2; i++){
        …
}
```



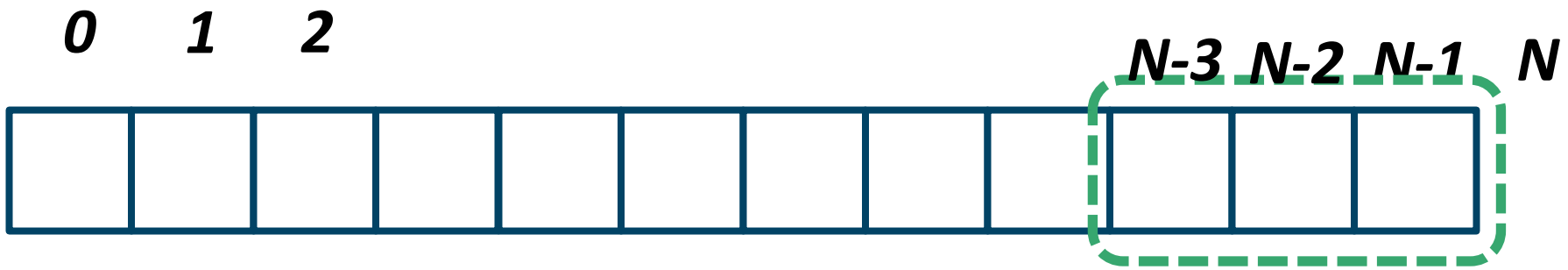
First 3

```
for (int i=0; i<3; i++){
        …
}
```

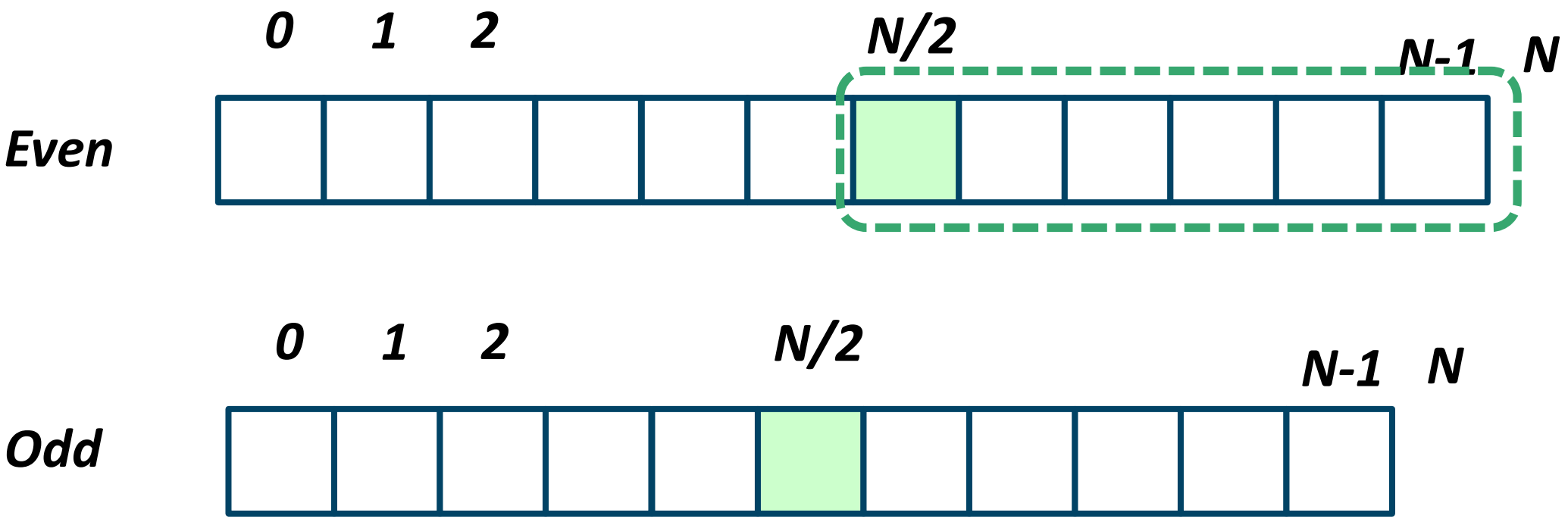


Last 3

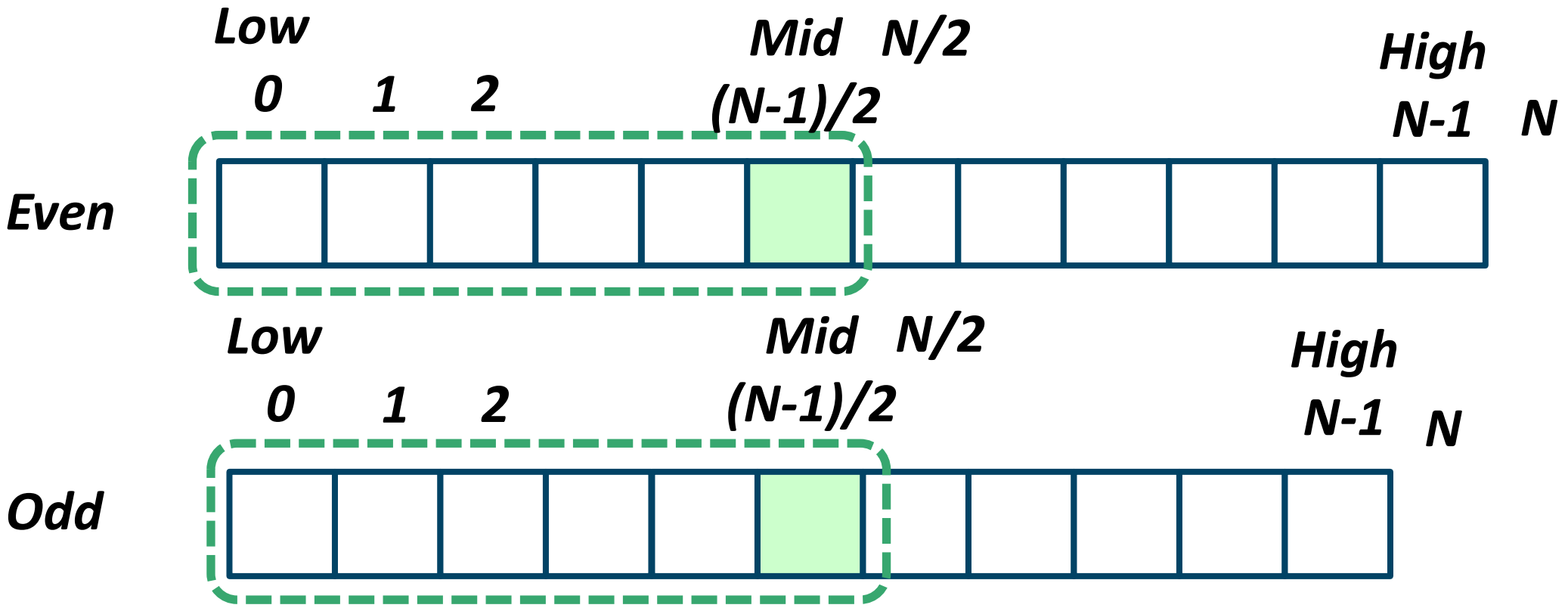
```
for (int i=N-3; i<N; i++){
        …
}
```



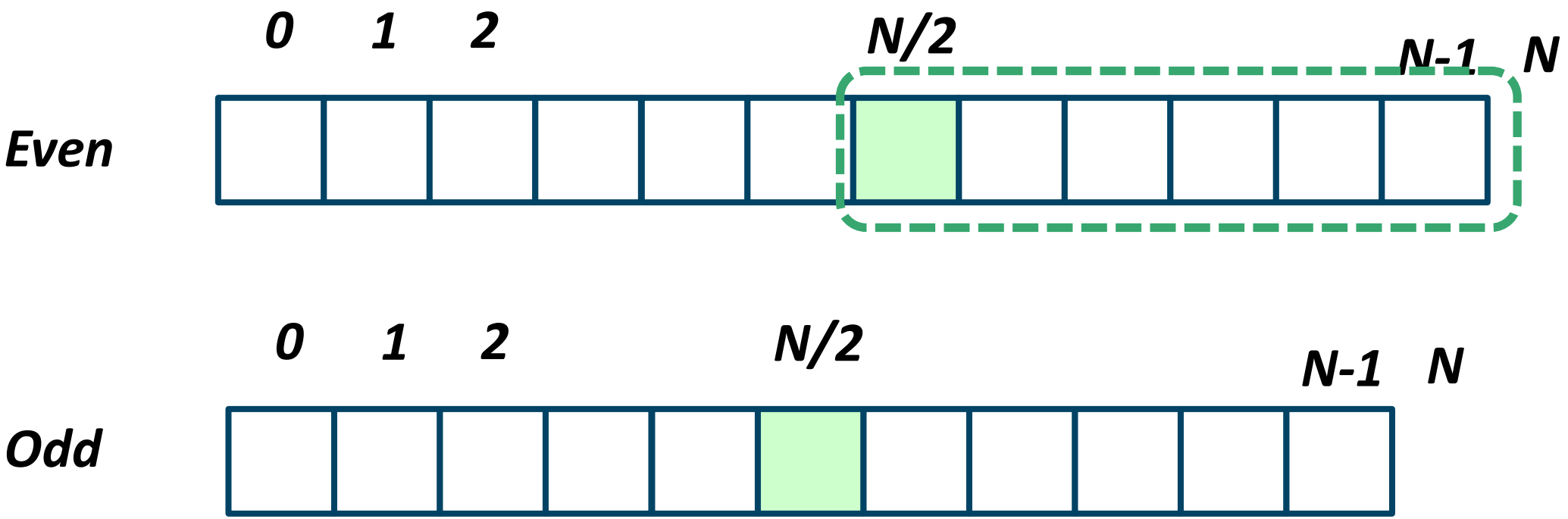
Middle by N/2



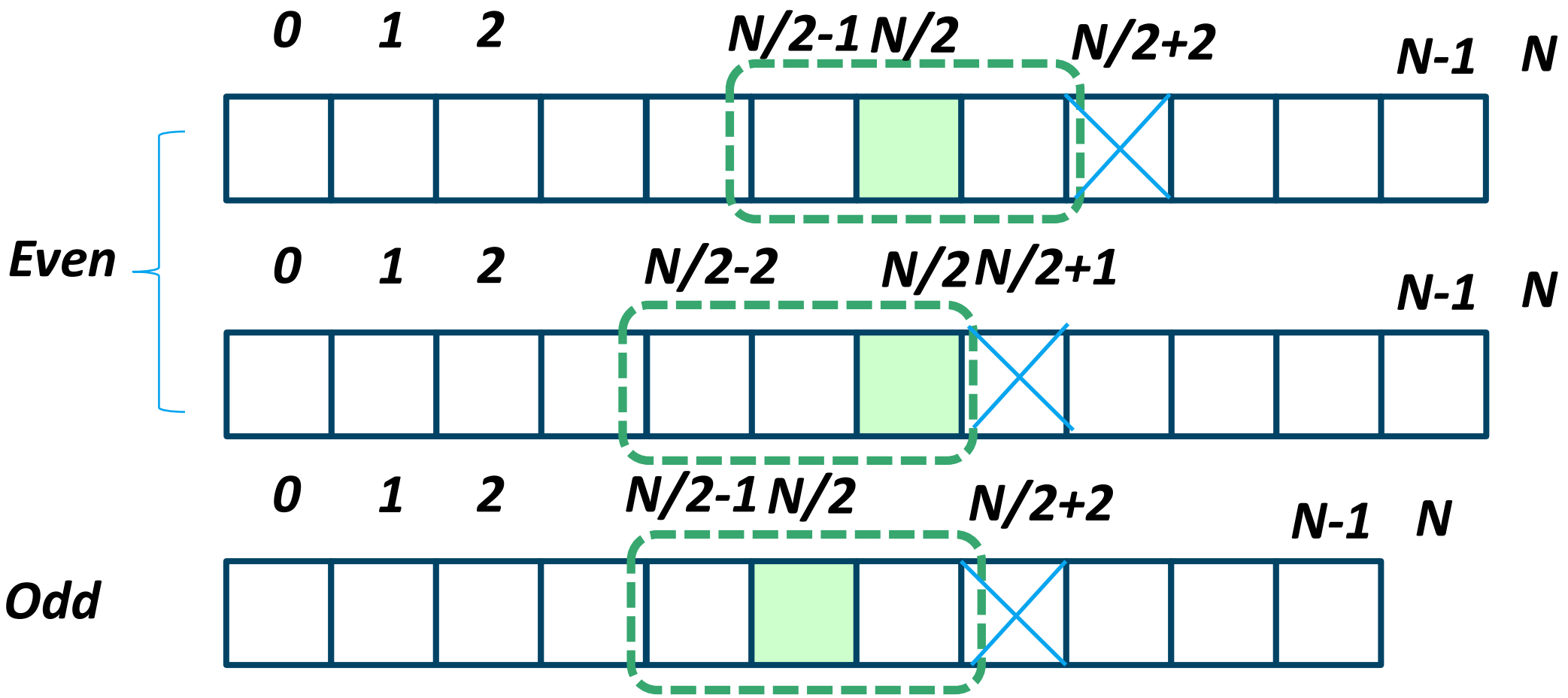
Middle by (low+high)/2



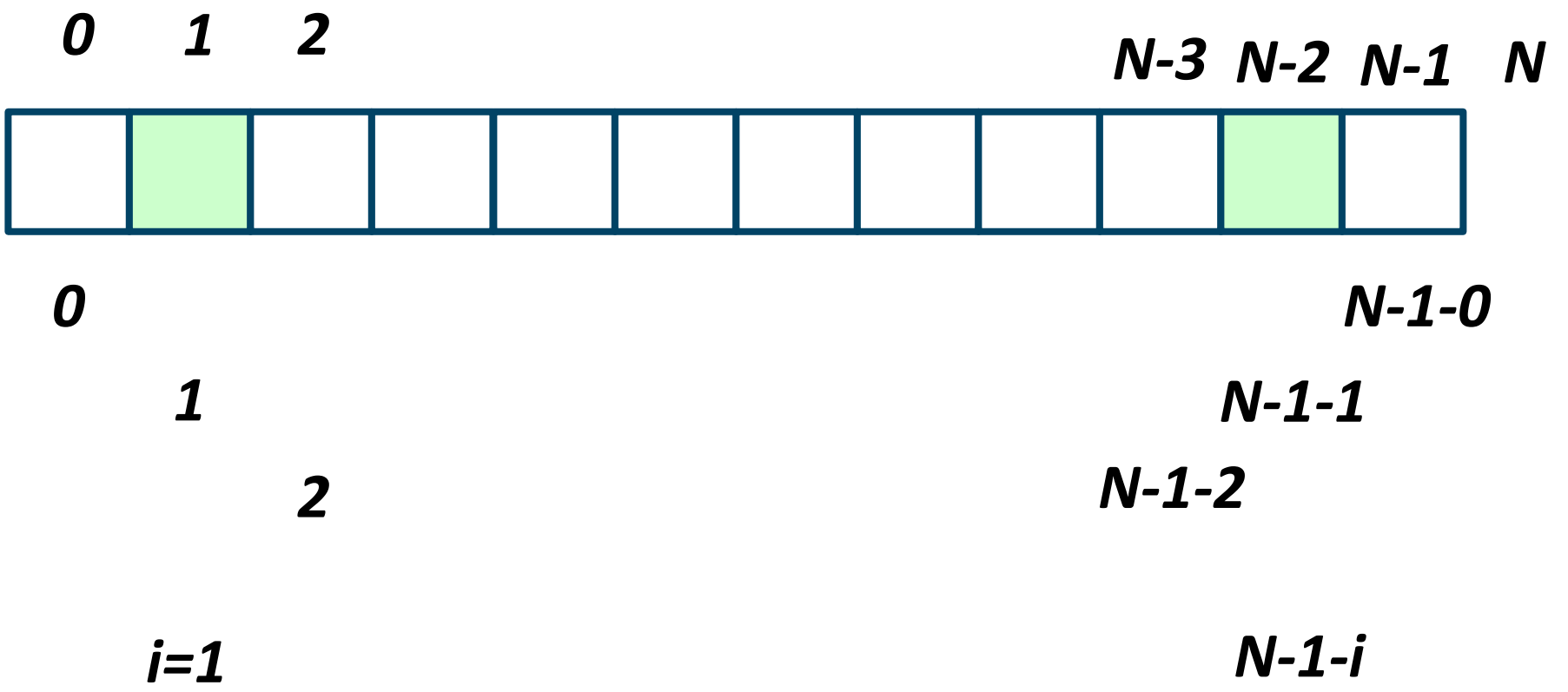
Middle by N/2

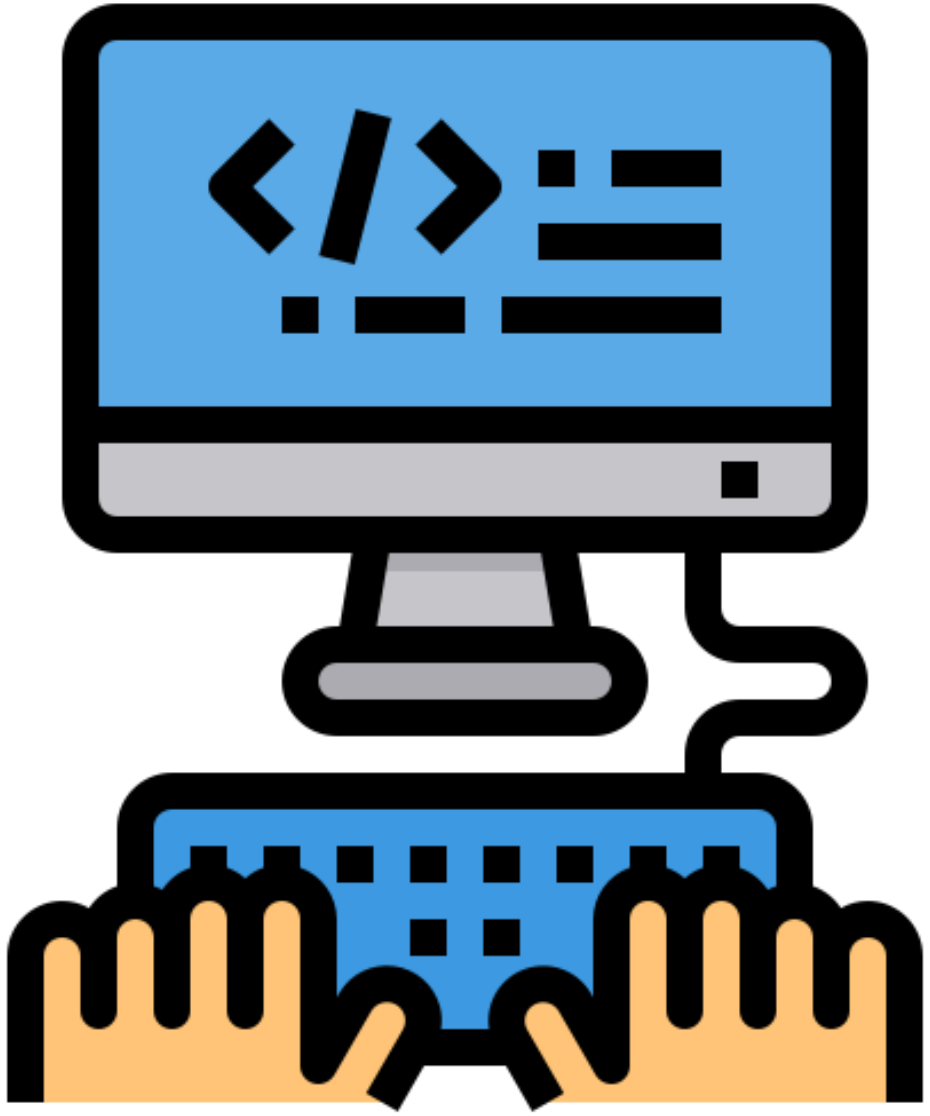


Middle 3



Symmetric Location





Array Processing

- Basic Algorithms

SECTION 6

Array Processing

- Min/Max/Sum
- Largest Index
- Shuffling
- Rotation
- Reverse
- Histogram (Doubloon, canSpell)

Sum

```
static int[] a = {2, 5, 3, 1, 4};
public static int sum(int[] a){
   int s = 0;
   for (int i=0; i<a.length; i++) s += a[i];
   return s;
}
```

Max 1

```
public static int max1(int[] a){
    int m = Integer.MIN_VALUE;
    for (int x: a){
        if (x>m) m = x;
    }
    return m;
}
```

Max 2

```
public static int max2(int[] a){
    int m = a[0];
    for (int i=1; i<a.length; i++){
        if (a[i] > m) m = a[i];
    }
    return m;
}
```

Min 1

```
public static int min1(int[] a){
    int m = Integer.MAX_VALUE;
    for (int x: a){
        if (x<m) m = x;
    }
    return m;
}
```

Min 2

```
public static int min2(int[] a){
    int m = a[0];
    for (int i=1; i<a.length; i++){
        if (a[i] < m) m = a[i];
    }
    return m;
}
```

Rotate Left

```
public static int[] rotateLeft(int[] a){
    int tmp = a[0];
    for (int i=1; i<a.length; i++){
        a[i-1] = a[i];
    }
    a[a.length-1] = tmp;
    return a;
}
```

Rotate Right

```
public static int[] rotateRight(int[] b){
    int tmp = b[b.length-1];
    for (int i=b.length-2; i>=0; i--){
        b[i+1] = b[i];
    }

    b[0] = tmp;
    return b;
}
```

Reverse 1

```
public static int[] reverse1(int[] a){
    int[] b = new int[a.length];
    for (int i=0; i<a.length; i++){
        b[a.length-1-i] = a[i];
    }
    return b;
}
```

Reverse 2

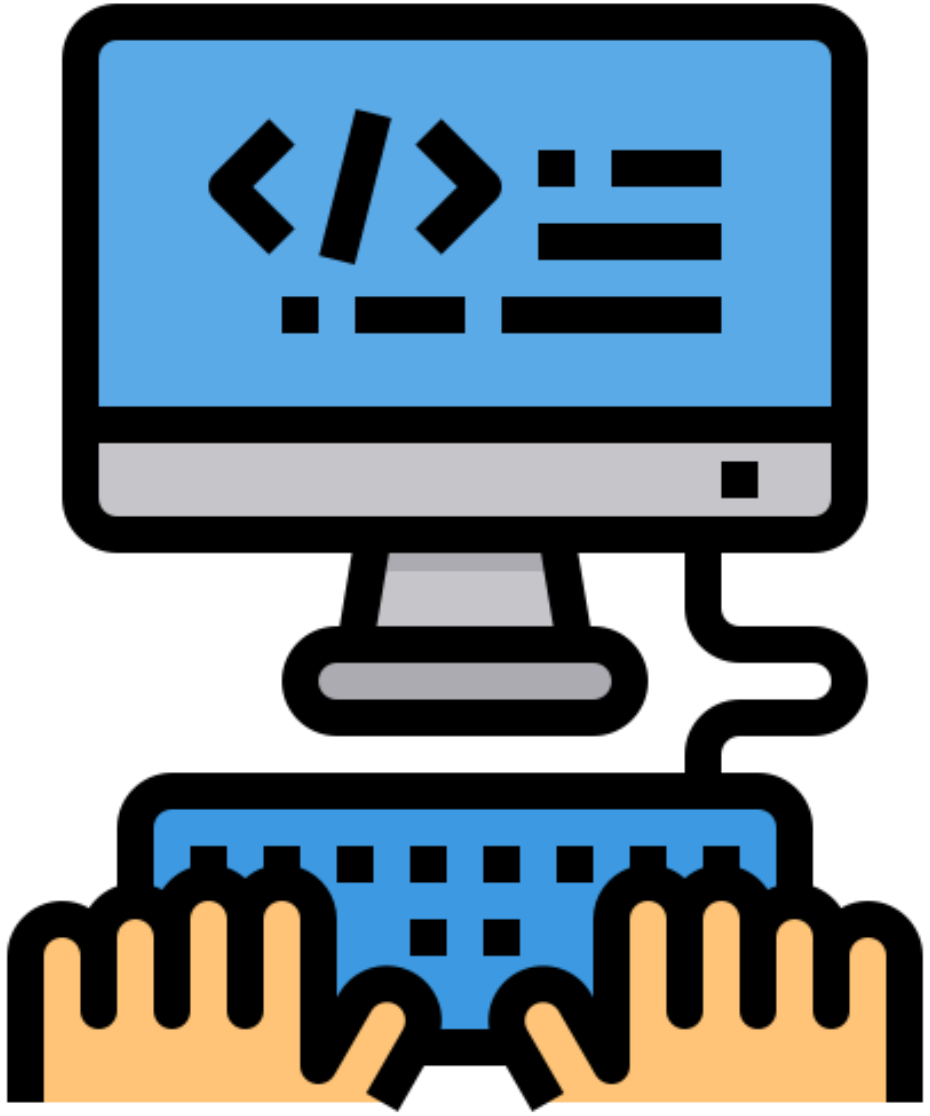
```
public static int[] reverse2(int[] a){
    for (int i=0; i<a.length/2; i++){
        int tmp = a[i];
        a[i] =a[a.length-1-i];
        a[a.length-1-i] = tmp;
    }
    return a;
}
```

```java
public class Histogram
{
    static String s = "abdjslkqiepqruqwoirafdafhdjznxadoauifqprqlsdjfhfhsdgajweyrw";

    public static int[] histogram(String s){
        int[] h = new int[26];
        for (char x : s.toCharArray()){
            h[x-'a']++;
        }
        return h;
    }

    public static void main(String[] args){
        int[] h = histogram(s);
        for (char x = 'a'; x <= 'z'; x++){
            System.out.printf("%c, %d\n", x, h[x-'a']);
        }
    }
}
```

eC Learning Channel



Practice MCQ Questions

SECTION 7

Multiple Choice (1)

What is the length of the array numbers after the following code executes:

```
String word = “Bear”;
int[] numbers = new int[word.length()-1];
```

Multiple Choice (1)

Ⓐ 1

Ⓑ 3

Ⓒ 4

Ⓓ The array will not be created due to a compile-time error.

Multiple Choice (1)

Ⓐ 1

Ⓑ 3

Ⓒ 4

Ⓓ The array will not be created due to a compile-time error.

Multiple Choice (2)

Consider the code segment:

```
String[] s = new String[2];
String[] t = {“Michael”, “Megan”,
“Chelsea”};
s = t;
System.out.println(s.length);
```

What is printed as a result fo executing the code segment?

Multiple Choice (2)

Ⓐ 2

Ⓑ 3

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be created due to a run-time error.

Multiple Choice (2)

Ⓐ 2

Ⓑ 3

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be created due to a run-time error.

Multiple Choice (3)

Consider the code segment:

```
final int[] a1 = {1, 2};
int[] b1 = {3, 4};
a1 = b1;
System.out.println(a1[1]);
```

What is printed as a result fo executing the code segment?

Multiple Choice (3)

Ⓐ 2

Ⓑ 4

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be created due to a run-time error.

Multiple Choice (3)

Ⓐ 2

Ⓑ 4

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be cre ated due to a run-time error.

Multiple Choice (3)

Ⓐ Ⅰ only

Ⓑ Ⅱ only

Ⓒ Ⅱ and Ⅲ

Ⓓ Ⅰ and Ⅱ

Multiple Choice (4)

Consider the code segment:

```
final int[] myArray = {1, 2};
myArray[1] = 3;
System.out.println(myArray[1]);
```

What is printed as a result fo executing the code segment?

Multiple Choice (4)

Ⓐ 2

Ⓑ 3

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be created due to a run-time error.

Multiple Choice (4)

Ⓐ 2

Ⓑ 3

Ⓒ Nothing will be printed due to a compile-time error

Ⓓ Nothing will not be created due to a run-time error.

Multiple Choice (5)

Consider the following incomplete method:

```
public static int mod3(int[] numbers){
   int count = 0;
   for (int i=0; i<numbers.length; i++){
      /* mystery code */
   }
   return count;
}
```

Multiple Choice (5)

Method mod3 is intended to return the number of integers int eh array numbers that are evenly divisible by 3. Which of the following code segments could be used to replace /* mystery code */ so that mod3 will work as intended?

```
I. if (i%3 == 0) { count++; }
II. if (numbers[i]%3 == 0) { count++; }
III. while (numbers[i]%3 == 0) { count++; }
```

Multiple Choice (5)

Ⓐ Ⅱ only

Ⓑ Ⅲ only.

Ⓒ Ⅰ and Ⅱ only.

Ⓓ Ⅱ and Ⅲ only.

Multiple Choice (6)

Assume that an array `arr` of integer values has been declared and initialized with random numbers. Which of the following code segments correctly swaps the values stored in `arr[0]` and `arr[3]`?

Multiple Choice (6)

Ⓐ
```
arr[0] = 3;
arr[3] = 0;
```

Ⓑ
```
arr[0] = arr[3];
arr[3] = arr[0];
```

Ⓒ
```
int k =  arr[3];
arr[3] = arr[0];
arr[0] = k;
```

Ⓓ
```
int k = arr[3];
arr[0] = arr[3];
arr[3] = k;
```

Multiple Choice (6)

Ⓐ
```
arr[0] = 3;
arr[3] = 0;
```

Ⓑ
```
arr[0] = arr[3];
arr[3] = arr[0];
```

Ⓒ
```
int k =  arr[3];
arr[3] = arr[0];
arr[0] = k;
```

Ⓓ
```
int k = arr[3];
arr[0] = arr[3];
arr[3] = k;
```

Multiple Choice (7)

Consider the code segment:

```
int[] scores = {75, 63, 52, 80};
for (int s: scores){
    if (s<75){
        s += 5;
    }
}
for (int s: scores) System.out.print(s+" ");
```

What is printed as a result fo executing the code segment?

Multiple Choice (7)

Ⓐ 75 63 52 80

Ⓑ 75 68 57 80

Ⓒ 80 68 57 80

Ⓓ Nothing will be printed due to a compile-time error.

Multiple Choice (7)

Ⓐ 75 63 52 80

Ⓑ 75 68 57 80

Ⓒ 80 68 57 80

Ⓓ Nothing will be printed due to a compile-time error.

Multiple Choice (8)

Consider the following code segment:

```
String[] names = {"Abi", "Brianna", "Jada",
      "Kiara", "Tiffany", "Vannessa"};
int middle = names.length/2 -1;
for (int x = 0; x<middle; x++){
   String hold = names[x];
   names[x] = names[names.length-x-1];
   names[names.length-x-1] = hold;
}
for (String n: names)
    System.out.print(n+" ");
```

What is printed as a result of executing the code segment?

Multiple Choice (8)

Ⓐ Abi Brianna Jada Kiara Tiffany Vennesa

Ⓑ Vennesa Tiffany Kiara Jada Brianna Abi

Ⓒ Vennesa Tiffany Jada Kiara Brianna Abi

Ⓓ Vennesa Brianna Jada Kiara Tiffany Abi

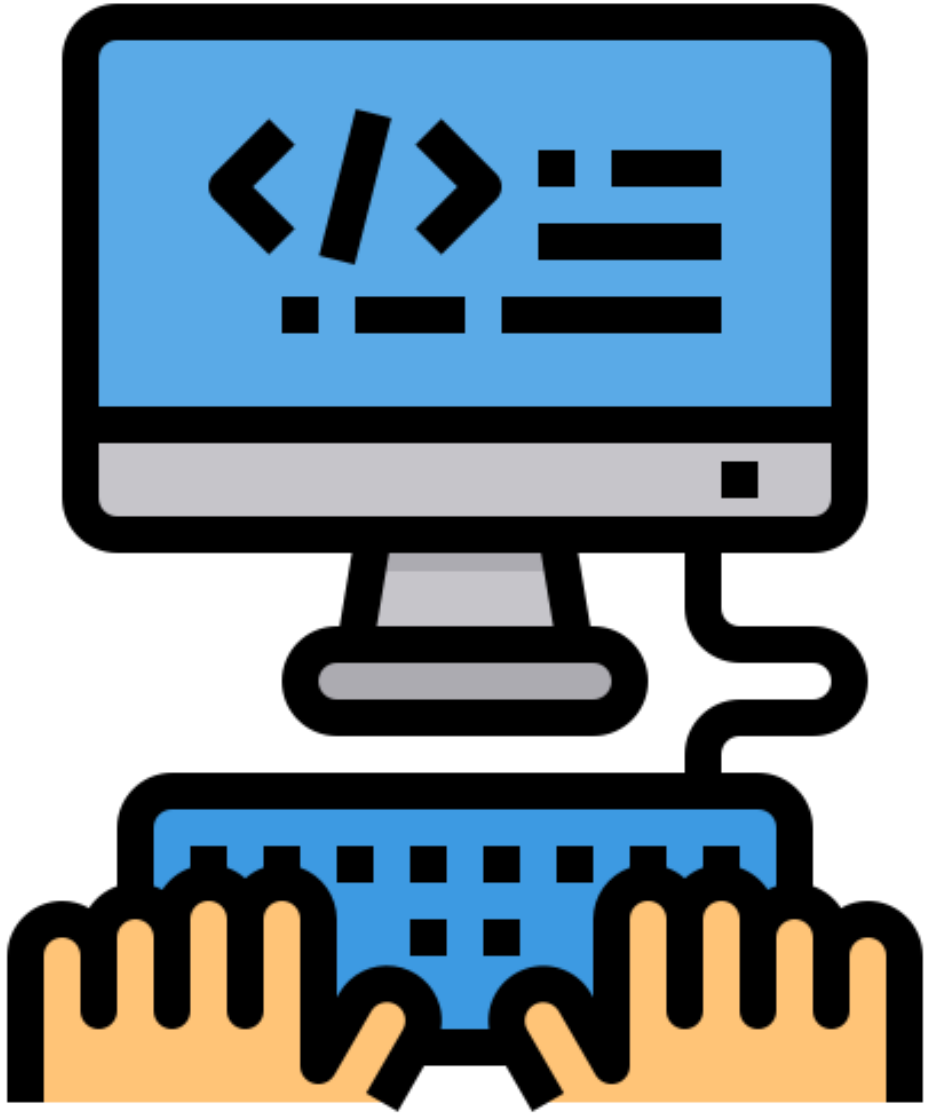
Multiple Choice (8)

Ⓐ Abi Brianna Jada Kiara Tiffany Vennesa

Ⓑ Vennesa Tiffany Kiara Jada Brianna Abi

Ⓒ Vennesa Tiffany Jada Kiara Brianna Abi

Ⓓ Vennesa Brianna Jada Kiara Tiffany Abi



Array Algorithms

SECTION 8

Histogram

```
public int[] getHistogram(String s){
    int[] h = new int[26];
    for (char c: s.toCharArray()){
        h[c-'A']++;
    }
    return h;
}
```

Linear Search

```
public int search(int[] a, int key){
    for (int i=0; i<a.length; i++){
      if (key==a[i]) return i;
    }
    return -1;
}
```

Insertion Sort

```
public int[] sort(int[] a){
  for (int i=1; i<a.length; i++){
    int j=i-1;
    int data = a[i];
    while (j>=0 && a[j]>data){
      a[j+1] = a[j]; j--;
    }
    j++;
    a[j] = data;
  }
  return a;
}
```

Selection Sort

```
public int[] sort(int[] a){
  for (int i=0; i<a.length-1; i++){
    int data = a[i];
    int idx = i;
    for (int j=i+1; j<a.length; j++){
      if(a[j]>a[idx]) idx = j;
    }
    if (idx!=i){
      int tmp = a[i];
      a[i] = a[idx];
      a[idx] = tmp;
    }
  }
  return a;
}
```

Bubble Sort

```
public int[] sort(int[] a){
  for (int i=a.length-1; i>0; i--){
    for (int j=0; j<i-1; j++){
      if (a[j]>a[j+1]){
        int tmp = a[j];
        a[j] = a[j+1];
        a[j+1] = tmp;
      }
    }
  }
  return a;
}
```

Swap Shuffle

```
public int[] shuffle(int[] a){
   for (int i=0; i<a.length; i++){
      int j= (int)(Math.random()*a.length);
      int k= a[i];
      a[i] = a[j];
      a[j] = k;
   }
   return a;
}
```

Zone Shuffle

```
public int[] shuffle(int[] a){
  for (int zone=a.length-1; zone>0; z--){
    int j= (int)(Math.random()*zone);
    int k=a[zone];
    a[zone] = a[j];
    a[j] = k;
  }
  return a;
}
```